5(a)	any **three** from: • greater bandwidth • does not suffer from (e.m.) interference / can be used in (e.m.) 'noisy' environments • no / less power / energy radiated / better security / less cross-talk • less attenuation / fewer repeaters / amplifiers needed • less weight / easier to handle / cheaper / occupy less space	**B3**
5(b)(i)	attenuation / gain = 10 log P_1 / P_2	**C1**
	$0.50 \times 57 = 10 \log (15 \times 10^{-3}/P)$ so $P = 2.1 \times 10^{-5}$ W *or* $-(0.50 \times 57) = 10 \log (P/15 \times 10^{-3})$ so $P = 2.1 \times 10^{-5}$ W	**A1**
5(b)(ii)	*either*	
	(calculation of S / N ratio at receiver) S / N ratio = $10 \log (2.1 \times 10^{-5} / 9.0 \times 10^{-7})$ or S/N ratio = 14	**M1**
	14 < 24 or S/N ratio < minimum S/N ratio	**A1**
	so not able to distinguish signal from noise	**A1**
	or	
	(calculation of minimum acceptable power at receiver) $24 = 10 \log (P / 9.0 \times 10^{-7})$ or $P = 2.3 \times 10^{-4}$	**(M1)**
	$2.1 \times 10^{-5} < 2.3 \times 10^{-4}$ or power < minimum power	**(A1)**
	so not able to distinguish signal from noise	**(A1)**

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